A Direct Proof of Liu's Entropy Bound for the Union-Closed Sets Conjecture

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Abstract

We prove a bivariate entropy inequality which yields the constant

$$c' = 0.382709087918735\ldots \tag{0.1}$$

for Frankl's union-closed sets conjecture. This constant is the one obtained by Liu through a conditionally independent coupling, subject to a numerical optimizer-structure hypothesis. Our proof replaces that hypothesis by a direct proof of the underlying entropy inequality. The argument combines a Boppana-style Rolle method on the diagonal, a degree-27 Bernstein positivity certificate, and a finite Taylor-model interval-arithmetic verification of the off-diagonal critical-point structure. As a consequence, every finite nonempty union-closed family different from $\{\varnothing\}$ has an element contained in at least a c'-fraction of its members. The code implementing all certificates is available at the repository listed in the appendix.

1 Introduction

A finite family $\mathcal{F}$ of sets is *union-closed* if

$$A, B \in \mathcal{F} \quad \Longrightarrow \quad A \cup B \in \mathcal{F}. \tag{1.1}$$

Frankl's union-closed sets conjecture, formulated in 1979, asserts that if $\mathcal{F}$ is finite, union-closed, and not equal to $\{\varnothing\}$, then some element belongs to at least half of the sets in $\mathcal{F}$ [1, 2]. Despite its elementary formulation, the conjecture has remained open for more than four decades. It is one of the central extremal problems concerning finite set systems.

A major breakthrough was made by Gilmer [3], who introduced an entropy method and proved the first absolute constant lower bound. In Gilmer's argument, one samples independent random sets A, B from $\mathcal{F}$, compares $H(A \cup B)$ with $H(A)$, and uses union-closedness to force a popular element. Gilmer's initial constant was 0.01. This was the first proof that there is an absolute $c > 0$ such that every finite nonempty union-closed family different from $\{\varnothing\}$ has an element appearing in at least a c-fraction of its members.

Shortly afterward, the constant was improved to

$$\frac{3 - \sqrt{5}}{2} = 0.3819660112\ldots \tag{1.2}$$

by several works, including Alweiss–Huang–Sellke [4], Chase–Lovett [5], Sawin [6], Cambie [8], Yu [9], and Pebody [10]. A central analytic ingredient in this line of work is the one-variable entropy inequality

$$h(x^2) \geq \varphi x h(x), \qquad 0 \leq x \leq 1, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \tag{1.3}$$

which goes back to Boppana's work on amplification of probabilistic Boolean formulas [11]. Chase and Lovett also showed that the constant $(3-\sqrt{5})/2$ is optimal for an approximate union-closed formulation, thereby identifying a barrier for the plain independent two-sample entropy method.

Liu [7] introduced a further refinement based on conditionally independent couplings. Instead of taking two independent samples, one allows the two samples to be independent only after conditioning on an auxiliary random variable. This produces a richer class of coordinatewise couplings and leads to the numerical constant

$$0.3827090879\ldots, \tag{1.4}$$

slightly above the golden-ratio threshold. Liu proved analytically a strict improvement over the previous 0.38234-level constants, but the explicit value $0.38271\ldots$ in his paper depends on additional numerically verified hypotheses. More precisely, Liu's explicit value requires two extra certificates: first, a positive-semidefinite condition for an infinite kernel matrix associated with the conditionally independent protocol; second, a global optimizer-structure hypothesis for a nine-dimensional variational problem. The optimizer is observed numerically to have a two-point boundary structure, but Liu does not prove this structure analytically.

The purpose of the present paper is to bypass both of these hypotheses. We do not prove Liu's nine-dimensional optimizer hypothesis, nor do we prove the infinite matrix positive-semidefinite condition in Liu's formulation. Instead, we prove directly the bivariate entropy inequality that acts as a dual certificate for the relevant coupling. Let

$$I(x,y) = xy(1+(1-x)(1-y)). \tag{1.5}$$

We prove that, for explicit constants m_* and β_*,

$$m_*\big((1-\beta_*)h(xy) + \beta_* h(I(x,y))\big) \geq \frac{xh(y)+yh(x)}{2} \qquad (0 \leq x, y \leq 1). \tag{1.6}$$

Here

$$m_* = 0.617290912081265\ldots, \qquad 1-m_* = 0.382709087918735\ldots. \tag{1.7}$$

The inequality (1.6) directly implies that every finite nonempty union-closed family different from $\{\varnothing\}$ has an element belonging to at least a

$$0.382709087918735\ldots \tag{1.8}$$

fraction of its sets.

The proof of (1.6) has three parts. On the boundary of the square $[0,1]^2$, the inequality is immediate from $m_* > 1/2$. On the diagonal $x = y$, we write $M(x) = G(x,x)$ and prove

$M(x) \geq 0$ by a fifth-derivative sign argument: a symbolic computation reduces the sign of $M^{(5)}$ to the positivity of a degree-27 polynomial P, and we certify $P > 0$ on $[0, 1]$ by verifying the positivity of all 28 of its Bernstein coefficients. Off the diagonal, any negative interior minimum would be an off-diagonal critical point. Passing to variables

$$u = \frac{x+y}{2}, \qquad v = \frac{x-y}{2},$$

we analyze the branch $F = \Gamma_v/v = 0$. Along this branch, the derivative of Γ_u is controlled by the Schur-complement identity

$$\frac{d}{du}\Gamma_u(u, v(u)) = \frac{\det D^2\Gamma(u, v(u))}{\Gamma_{vv}(u, v(u))}.$$

A Taylor-model interval computation certifies the required determinant signs on the two branch arms, while a separate interval-Newton and topological argument certifies that there are no additional components of $F = 0$. This rules out all off-diagonal critical points.

Computer-assisted inequalities have already played a role in the recent entropy approach to the union-closed conjecture. In particular, Alweiss–Huang–Sellke used computer verification in their treatment of the one-variable inequality underlying the $(3 - \sqrt{5})/2$ bound [4]. The present paper uses certified computation in the same spirit: all numerical claims are reduced to finite interval statements with explicit margins. The analytic part of the proof identifies the correct bivariate inequality and reduces the off-diagonal problem to branch isolation and determinant sign checks; the appendix records the finite certificates needed to make those reductions rigorous.

2 Definitions and constants

Throughout the paper we use natural logarithms. Define the binary entropy function by

$$h(t) = -t\log t - (1-t)\log(1-t), \qquad 0 \leq t \leq 1, \tag{2.1}$$

with the convention $0 \log 0 = 0$. Let

$$L(t) = h'(t) = \log\frac{1-t}{t}. \tag{2.2}$$

Define

$$I(x, y) = xy(1 + (1-x)(1-y)) = xy(2 - x - y + xy), \tag{2.3}$$

and

$$D(x) = I(x, x) = x^2(2 - 2x + x^2). \tag{2.4}$$

Let $x_* \in (0, 1)$ be the unique root of

$$x^4 - 2x^3 + 3x^2 - 1 = 0. \tag{2.5}$$

Equivalently,

$$x_*^2(2 + (1 - x_*)^2) = 1. \tag{2.6}$$

Set
$$p_* = \frac{h(x_*)}{h(x_*^2)}, \tag{2.7}$$
$$m_* = p_* x_*, \qquad c' = 1 - m_*. \tag{2.8}$$
Numerically,
$$x_* = 0.690787593924988\ldots, \qquad m_* = 0.617290912081265\ldots, \qquad c' = 0.382709087918735\ldots. \tag{2.9}$$

The parameter β_* is defined by the stationarity condition
$$\frac{d}{dx}\left[m_*\big((1-\beta_*)h(x^2) + \beta_* h(D(x))\big) - xh(x)\right]_{x=x_*} = 0. \tag{2.10}$$
Equivalently,
$$\beta_* = \frac{\dfrac{h(x_*^2)}{h(x_*)}\left(h'(x_*) + \dfrac{h(x_*)}{x_*}\right) - 2x_* h'(x_*^2)}{D'(x_*)h'(D(x_*)) - 2x_* h'(x_*^2)}. \tag{2.11}$$
Numerically,
$$\beta_* = 0.100052559862893\ldots. \tag{2.12}$$
We write
$$\alpha_* = 1 - \beta_*. \tag{2.13}$$

Define
$$G(x,y) = m_*\big(\alpha_* h(xy) + \beta_* h(I(x,y))\big) - \frac{xh(y) + yh(x)}{2}. \tag{2.14}$$
The main analytic result of the paper is the inequality
$$G(x,y) \geq 0 \qquad (0 \leq x, y \leq 1). \tag{2.15}$$

We record the equality identities at (x_*, x_*). From (2.5) and (2.4),
$$D(x_*) = 1 - x_*^2. \tag{2.16}$$
By entropy symmetry,
$$h(D(x_*)) = h(1 - x_*^2) = h(x_*^2). \tag{2.17}$$
Therefore
$$G(x_*, x_*) = m_* h(x_*^2) - x_* h(x_*) = 0, \tag{2.18}$$
where the last equality is (2.7) and (2.8). Finally, (2.10) says
$$\frac{d}{dx}G(x,x)\bigg|_{x=x_*} = 0. \tag{2.19}$$
Since G is symmetric, (2.19) implies
$$\nabla G(x_*, x_*) = 0. \tag{2.20}$$

3 Boundary cases

We first prove nonnegativity on the boundary of the square. If $x = 0$, then $xy = 0$, $I(0, y) = 0$, and $xh(y) = 0$, hence

$$G(0, y) = 0. \tag{3.1}$$

Similarly,

$$G(x, 0) = 0. \tag{3.2}$$

If $x = 1$, then $xy = y$ and $I(1, y) = y$, so

$$G(1, y) = \left(m_* - \frac{1}{2}\right) h(y) \geq 0, \tag{3.3}$$

because $m_* > 1/2$. By symmetry,

$$G(x, 1) = \left(m_* - \frac{1}{2}\right) h(x) \geq 0. \tag{3.4}$$

Thus $G \geq 0$ on $\partial[0, 1]^2$, with equality on the axes and at the corner $(1, 1)$.

4 The diagonal inequality

Define

$$M(x) = G(x, x). \tag{4.1}$$

Using $I(x, x) = D(x)$, we have

$$M(x) = m_*\big(\alpha_* h(x^2) + \beta_* h(D(x))\big) - xh(x). \tag{4.2}$$

Lemma 1 (Diagonal nonnegativity)**.** *For every $x \in [0, 1]$,*

$$M(x) \geq 0. \tag{4.3}$$

On $[m_*, 1]$, equality occurs only at $x = x_*$ and $x = 1$. On $[0, 1]$, equality occurs at $x = 0$, $x = x_*$, and $x = 1$.

Proof. A symbolic differentiation gives

$$M^{(5)}(x) = -\frac{2P(x)}{x^3(1-x)^4(1+x)^4(x^2-2x+2)^4(x^3-x^2+x+1)^4}. \tag{4.4}$$

The denominator is strictly positive on $(0, 1)$. The polynomial P is

$$P(x) = P_0(x) + m_* P_1(x) + m_* \beta_* P_2(x). \tag{4.5}$$

The three polynomials are

$$\begin{aligned} P_0(x) = {} & 3x^{27} - 36x^{26} + 200x^{25} - 666x^{24} + 1364x^{23} - 1360x^{22} - 1096x^{21} + 6288x^{20} \\ & - 9854x^{19} + 4312x^{18} + 11216x^{17} - 23492x^{16} + 15180x^{15} + 11376x^{14} - 29832x^{13} \end{aligned}$$

$$+ 18608x^{12} + 9251x^{11} - 22292x^{10} + 9976x^9 + 6670x^8 - 9176x^7 + 1408x^6 + 2784x^5 - 1488x^4 - 400x^3 + 320x^2 - 32, \tag{4.6}$$

$$\begin{aligned} P_1(x) = {} & 24x^{26} - 288x^{25} + 1860x^{24} - 8208x^{23} + 27296x^{22} - 71472x^{21} + 150320x^{20} \\ & - 254064x^{19} + 338288x^{18} - 333680x^{17} + 197272x^{16} + 27248x^{15} - 209888x^{14} \\ & + 233136x^{13} - 104048x^{12} - 45904x^{11} + 96120x^{10} - 46640x^9 - 12924x^8 \\ & + 25312x^7 - 6336x^6 - 5312x^5 + 3360x^4 + 640x^3 - 640x^2 + 64, \end{aligned} \tag{4.7}$$

$$\begin{aligned} P_2(x) = {} & 24x^{26} - 168x^{25} + 60x^{24} + 2832x^{23} - 14144x^{22} + 41712x^{21} - 99488x^{20} \\ & + 213456x^{19} - 389072x^{18} + 532880x^{17} - 458888x^{16} + 88048x^{15} + 372160x^{14} \\ & - 587200x^{13} + 435712x^{12} - 83408x^{11} - 223240x^{10} + 324792x^9 - 210324x^8 \\ & + 22720x^7 + 77632x^6 - 70096x^5 + 29760x^4 - 5760x^3 - 256x^2 + 192x + 64. \end{aligned} \tag{4.8}$$

The degree-27 Bernstein coefficients b_i of P on $[0, 1]$ have certified lower bounds shown in Table 1. Since the Bernstein basis is nonnegative on $[0, 1]$, the table implies

$$P(x) > 0 \qquad (0 \le x \le 1). \tag{4.9}$$

Consequently,

$$M^{(5)}(x) < 0 \qquad (0 < x < 1). \tag{4.10}$$

Table 1: Certified lower bounds for the Bernstein coefficients of P.

i	lower bound	i	lower bound
0	11.45935667296576	14	4.610838565116360
1	11.898549817384106	15	4.964896105425193
2	12.078833739776622	16	5.668547978061419
3	11.876898948434980	17	6.798303977430921
4	11.307562491478990	18	8.465673305310538
5	10.464009676814568	19	10.840728674923076
6	9.465777139811940	20	14.191030026624640
7	8.423319403297192	21	18.945974671912115
8	7.421560841110654	22	25.804885986190340
9	6.518660405060565	23	35.924168951745940
10	5.753064074484635	24	51.256141180419560
11	5.152687277174934	25	75.205102681605100
12	4.742718854790539	26	114.019777429866760
13	4.550899799454681	27	180.158840956815370

We first prove the claim on $[m_*, 1]$. The following signs are certified by interval evaluation:

$$M(m_*) > 0, \qquad M'(m_*) < 0, \qquad M''(m_*) > 0,$$

$$M^{(3)}(m_*) > 0, \qquad M^{(4)}(m_*) < 0. \tag{4.11}$$

Numerically these are

$$\begin{aligned} M(m_*) &= 0.000777106330304900\ldots, \\ M'(m_*) &= -0.0199407396705102\ldots, \\ M''(m_*) &= 0.219173887997792\ldots, \\ M^{(3)}(m_*) &= 1.55739338128487\ldots, \\ M^{(4)}(m_*) &= -4.41104974811261\ldots. \end{aligned} \tag{4.12}$$

Moreover,

$$M(x_*) = 0, \qquad M'(x_*) = 0, \tag{4.13}$$

and interval evaluation gives

$$M''(x_*) = 0.317062949668787\ldots > 0. \tag{4.14}$$

We also need the behavior at 1. Put $s = 1 - x$. As $s \downarrow 0$,

$$\begin{aligned} h(x^2) &= 2s \log \frac{1}{s} + O(s), \\ h(D(x)) &= 2s \log \frac{1}{s} + O(s), \\ xh(x) &= s \log \frac{1}{s} + O(s). \end{aligned} \tag{4.15}$$

Therefore

$$M(x) = (2m_* - 1)s \log \frac{1}{s} + O(s). \tag{4.16}$$

Since $2m_* - 1 > 0$, differentiating with respect to $x = 1 - s$ gives

$$\begin{aligned} M'(1^-) &= -\infty, \\ M''(1^-) &= -\infty, \\ M^{(3)}(1^-) &= -\infty, \\ M^{(4)}(1^-) &= -\infty. \end{aligned} \tag{4.17}$$

By (4.10), $M^{(4)}$ is strictly decreasing. Since $M^{(4)}(m_*) < 0$, we have

$$M^{(4)}(x) < 0 \qquad (m_* \le x < 1). \tag{4.18}$$

Hence $M^{(3)}$ is strictly decreasing. From $M^{(3)}(m_*) > 0$ and $M^{(3)}(1^-) = -\infty$, there is a unique

$$a \in (m_*, 1) \tag{4.19}$$

with $M^{(3)}(a) = 0$. Thus M'' increases on (m_*, a) and decreases on $(a, 1)$. Since $M''(m_*) > 0$ and $M''(1^-) = -\infty$, there is a unique

$$b \in (a, 1) \tag{4.20}$$

with $M''(b) = 0$.

Consequently M' increases on (m_*, b) and decreases on $(b, 1)$. We have $M'(m_*) < 0$, $M'(x_*) = 0$, and $M''(x_*) > 0$, so $x_* \in (m_*, b)$ and is the unique zero of M' in (m_*, b). Also $M'(b) > 0$, while $M'(1^-) = -\infty$; hence there is a unique

$$d \in (b, 1) \tag{4.21}$$

with $M'(d) = 0$. The sign pattern is

$$M' < 0 \text{ on } (m_*, x_*), \qquad M' > 0 \text{ on } (x_*, d), \qquad M' < 0 \text{ on } (d, 1). \tag{4.22}$$

Since $M(m_*) > 0$, $M(x_*) = 0$, and $M(1) = 0$, it follows that

$$M(x) \geq 0 \qquad (m_* \leq x \leq 1). \tag{4.23}$$

It remains to treat $[0, m_*]$. As $x \downarrow 0$,

$$\begin{aligned} h(x^2) &= 2x^2 \log \frac{1}{x} + O(x^2), \\ h(D(x)) &= 4x^2 \log \frac{1}{x} + O(x^2), \\ xh(x) &= x^2 \log \frac{1}{x} + O(x^2). \end{aligned} \tag{4.24}$$

Hence

$$M(x) = \gamma x^2 \log \frac{1}{x} + O(x^2), \qquad \gamma = 2m_*(1 + \beta_*) - 1 > 0. \tag{4.25}$$

Thus $M(x) > 0$ for all sufficiently small positive x, $M'(0^+) = 0$, $M''(0^+) = +\infty$, $M^{(3)}(0^+) = -\infty$, and $M^{(4)}(0^+) = +\infty$.

Since $M^{(5)} < 0$, $M^{(4)}$ is strictly decreasing on $(0, m_*)$. From $M^{(4)}(0^+) = +\infty$ and $M^{(4)}(m_*) < 0$, there is a unique zero $c \in (0, m_*)$ of $M^{(4)}$. Thus $M^{(3)}$ increases on $(0, c)$ and decreases on (c, m_*). Since $M^{(3)}(0^+) = -\infty$ and $M^{(3)}(m_*) > 0$, $M^{(3)}$ has exactly one zero $a_0 \in (0, c)$. Therefore M'' decreases on $(0, a_0)$ and increases on (a_0, m_*).

The interval evaluation

$$M''(0.1) < 0 \tag{4.26}$$

combined with $M''(0^+) = +\infty$ and $M''(m_*) > 0$ implies that M'' has exactly two zeros $b_1 < b_2$ in $(0, m_*)$. Thus M' increases, then decreases, then increases. Since $M'(0^+) = 0$ and $M'(x) > 0$ for small $x > 0$, while $M'(m_*) < 0$, the function M' has exactly one zero $r \in (0, m_*)$. Hence M increases on $(0, r)$ and decreases on (r, m_*). Since $M(0) = 0$ and $M(m_*) > 0$, we get

$$M(x) \geq 0 \qquad (0 \leq x \leq m_*). \tag{4.27}$$

This completes the proof. $\square$

5 Off-diagonal critical points

Introduce
$$u = \frac{x+y}{2}, \qquad v = \frac{x-y}{2}. \tag{5.1}$$
Thus
$$x = u + v, \qquad y = u - v. \tag{5.2}$$
The feasible half-triangle is
$$0 < u < 1, \qquad 0 < v < \min(u, 1-u). \tag{5.3}$$
Define
$$\Gamma(u, v) = G(u+v, u-v). \tag{5.4}$$
Since G is symmetric, Γ is even in v. For $v > 0$, define
$$F(u, v) = \frac{\Gamma_v(u, v)}{v}. \tag{5.5}$$
An off-diagonal critical point satisfies
$$F(u, v) = 0, \qquad \Gamma_u(u, v) = 0, \qquad v > 0. \tag{5.6}$$

Set
$$a = u^2 - v^2, \qquad b = 2 - 2u + u^2 - v^2, \qquad J = ab. \tag{5.7}$$
Thus $a = xy$, $b = 1 + (1-x)(1-y)$, and $J = I(x, y)$. Let
$$\Delta(u, v) = \det D^2\Gamma(u, v) = \Gamma_{uu}\Gamma_{vv} - \Gamma_{uv}^2. \tag{5.8}$$
Along a regular branch $v = v(u)$ of $F = 0$, one has $\Gamma_v(u, v(u)) = 0$. Differentiating gives
$$v'(u) = -\frac{\Gamma_{uv}}{\Gamma_{vv}}. \tag{5.9}$$
For
$$\Phi(u) = \Gamma_u(u, v(u)), \tag{5.10}$$
we therefore get
$$\Phi'(u) = \Gamma_{uu} - \frac{\Gamma_{uv}^2}{\Gamma_{vv}} = \frac{\Delta}{\Gamma_{vv}}. \tag{5.11}$$
This identity is the monotonicity mechanism for the off-diagonal analysis.

5.1 Endpoint positivity on the upper arc

Put

$$x = 1 - \varepsilon, \qquad y = s. \tag{5.12}$$

Let

$$q = 2m_* - 1 = 0.234581824162530\ldots. \tag{5.13}$$

On the upper branch, $F = 0$ is equivalent to $G_x = G_y$, hence

$$\Gamma_u = G_x + G_y = 2G_y. \tag{5.14}$$

Since

$$G_y = m_*\big(\alpha_* x L(xy) + \beta_* I_y(x,y) L(I(x,y))\big) - \frac{h(x) + xL(y)}{2}, \tag{5.15}$$

we have the exact identity

$$\Gamma_u = 2m_*\big(\alpha_* x L(xy) + \beta_* I_y(x,y) L(I(x,y))\big) - h(x) - xL(y). \tag{5.16}$$

Here

$$I_y(x,y) = x(2 - x - 2y + 2xy). \tag{5.17}$$

Lemma 2 (Endpoint positivity)**.** *For every point of the upper arc with $0 < s = y \le s_1 = 0.04$,*

$$\Gamma_u \ge \frac{2m_* - 1}{2} \log \frac{1}{s} > 0. \tag{5.18}$$

Proof. The endpoint branch satisfies

$$0 < \varepsilon(s) \le s^2 \qquad (0 < s \le 0.04). \tag{5.19}$$

Indeed, from the explicit formula for $F = (G_x - G_y)/v$, one obtains

$$\begin{aligned} &\lim_{\varepsilon \downarrow 0} F(\varepsilon, s) = +\infty, \\ &F(s^2, s) \le -\frac{q}{2} \log \frac{1}{s} + 0.003 < 0, \end{aligned} \tag{5.20}$$

and

$$\partial_\varepsilon F(\varepsilon, s) \le -\frac{1}{4s} < 0 \qquad (0 < \varepsilon \le s^2,\ 0 < s \le 0.04). \tag{5.21}$$

Thus the unique endpoint solution satisfies (5.19).

Assume $0 < s \le s_1$ and $0 < \varepsilon \le s^2$. Then $x = 1 - \varepsilon \ge 1 - s^2$. Since $xy \le s$ and L is decreasing,

$$L(xy) \ge L(s). \tag{5.22}$$

Moreover,

$$I_y = (1 - \varepsilon)(1 + \varepsilon - 2\varepsilon s) \ge 1 - s^2, \tag{5.23}$$

and

$$I(x,y) \le s(1 + s^2). \tag{5.24}$$

Therefore

$$L(I(x,y)) \geq L(s(1+s^2)). \tag{5.25}$$

Also $h(x) = h(\varepsilon) \leq h(s^2)$ and $-xL(y) \geq -L(s)$. Substituting these bounds into (5.16) gives

$$\Gamma_u \geq 2m_* \big(\alpha_*(1-s^2)L(s) + \beta_*(1-s^2)L(s(1+s^2))\big) - h(s^2) - L(s). \tag{5.26}$$

Let

$$\delta(s) = L(s) - L(s(1+s^2)). \tag{5.27}$$

Then

$$\Gamma_u \geq \big(2m_*(1-s^2) - 1\big)L(s) - 2m_*\beta_*(1-s^2)\delta(s) - h(s^2). \tag{5.28}$$

For $s \leq 1/25$,

$$L(s) \geq \log\frac{1}{s} - \frac{1}{24}, \tag{5.29}$$

while

$$\delta(s) \leq 2s^2, \tag{5.30}$$

and

$$h(s^2) \leq h(s_1^2). \tag{5.31}$$

Set

$$A_0 = 2m_*(1-s_1^2) - 1, \qquad C_0 = \frac{A_0}{24} + 4m_*\beta_* s_1^2 + h(s_1^2). \tag{5.32}$$

Numerically,

$$A_0 = 0.232606493243870\ldots, \qquad C_0 = 0.0219863330048360\ldots. \tag{5.33}$$

Thus

$$\Gamma_u \geq A_0 \log\frac{1}{s} - C_0. \tag{5.34}$$

Finally,

$$\Big(A_0 - \frac{q}{2}\Big)\log 25 - C_0 = 0.349200203430100\ldots > 0. \tag{5.35}$$

Since $\log(1/s) \geq \log 25$, (5.18) follows. □

Proposition 3 (Certified branch structure)**.** *In the feasible region (5.3), the zero set $F(u,v) = 0$ consists of exactly two smooth arcs meeting at a single fold point. More precisely:*

(i) The lower arc begins at the bifurcation point $(x_0, 0)$, with

$$x_0 = 0.220715379651680\ldots, \tag{5.36}$$

and ends at the fold

$$(u_f, v_f) = (0.552276058360800\ldots, 0.434687834510229\ldots). \tag{5.37}$$

Along this arc,

$$\Gamma_{vv} < 0, \qquad \Delta < 0. \tag{5.38}$$

(ii) *The upper arc begins at the boundary point* $(x, y) = (1, 0)$ *and ends at* (u_f, v_f)*. On the compact part* $y \geq 0.04$*,*

$$\Gamma_{vv} > 0, \qquad \Delta < 0. \tag{5.39}$$

For $0 < y \leq 0.04$*, Lemma 2 gives* $\Gamma_u > 0$*.*

(iii) *The endpoint values satisfy*

$$\Gamma_u(x_0, 0) > 0, \qquad \Gamma_u(u_f, v_f) > 0.41. \tag{5.40}$$

Proof. The proposition is the certified interval statement proved in Appendix A. The proof uses one-variable interval Newton for the bifurcation point, interval Newton for the fold, Taylor-model interval arithmetic for (5.38) and (5.39), and a uniform band certificate for the approach to $(1, 0)$. □

Lemma 4 (No off-diagonal critical points)**.** *There is no point* $(x, y) \in (0, 1)^2$ *with* $x \neq y$ *such that*

$$G_x(x, y) = G_y(x, y) = 0. \tag{5.41}$$

Proof. By symmetry, assume $v > 0$. Any such critical point must satisfy $F = 0$ and $\Gamma_u = 0$. On the lower arc, Proposition 3 gives $\Gamma_{vv} < 0$ and $\Delta < 0$. Hence by (5.11), Γ_u is strictly increasing along the lower arc. Since $\Gamma_u(x_0, 0) > 0$, $\Gamma_u > 0$ throughout the lower arc.

On the upper arc with $y \leq 0.04$, Lemma 2 gives $\Gamma_u > 0$. On the compact part of the upper arc, Proposition 3 gives $\Gamma_{vv} > 0$ and $\Delta < 0$. Hence by (5.11), Γ_u decreases toward the fold. Since $\Gamma_u(u_f, v_f) > 0.41$, it remains positive on this compact part. Therefore no point on $F = 0$, $v > 0$, can satisfy $\Gamma_u = 0$. □

6 Proof of the main inequality

Theorem 5 (Bivariate entropy inequality)**.** *For every* $x, y \in [0, 1]$*,*

$$m_*\big((1 - \beta_*)h(xy) + \beta_* h(I(x, y))\big) \geq \frac{xh(y) + yh(x)}{2}. \tag{6.1}$$

Equality occurs on the axes, at (x_*, x_*)*, and at* $(1, 1)$*.*

Proof. This is exactly the assertion $G(x, y) \geq 0$. Since G is continuous on $[0, 1]^2$, it attains a minimum. If a minimizer lies on the boundary, Section 3 gives nonnegativity. If it lies in the interior and on the diagonal, Lemma 1 gives nonnegativity. If it lies in the interior and off the diagonal, it would be an off-diagonal critical point, contradicting Lemma 4. Thus $G \geq 0$ everywhere. The equality cases are the boundary equalities from Section 3 and the diagonal equality (x_*, x_*) from Lemma 1. □

7 The union-closed corollary

Corollary 6. *Let $\mathcal{F} \subseteq 2^{[n]}$ be finite, nonempty, union-closed, and not equal to $\{\varnothing\}$. Then some element belongs to at least*

$$c' = 1 - m_* = 0.382709087918735\ldots \tag{7.1}$$

of the sets in $\mathcal{F}$.

Proof. Let A be uniform on $\mathcal{F}$. For a set S, write

$$Z_i(S) = \mathbf{1}_{\{i \notin S\}}. \tag{7.2}$$

Assume for contradiction that every element belongs to fewer than a c'-fraction of the sets. Equivalently,

$$\mathbb{P}[i \notin A] > 1 - c' = m_* \tag{7.3}$$

for every coordinate i.

Reveal coordinates in order and define

$$X_i = \mathbb{P}[Z_i(A) = 1 \mid Z_1(A), \ldots, Z_{i-1}(A)]. \tag{7.4}$$

Then the chain rule gives

$$H(A) = \sum_{i=1}^{n} \mathbb{E}h(X_i), \tag{7.5}$$

and

$$\mathbb{E}X_i = \mathbb{P}[i \notin A] > m_*. \tag{7.6}$$

We use two coordinatewise couplings. First, if the conditional absence probabilities for two histories are x and y, the independent coupling gives absence probability xy in the union. Second, define a coupling of two absence bits (U, V) by

$$\begin{aligned}
\mathbb{P}[U = 1, V = 1] &= I(x, y), \\
\mathbb{P}[U = 1, V = 0] &= x - I(x, y), \\
\mathbb{P}[U = 0, V = 1] &= y - I(x, y), \\
\mathbb{P}[U = 0, V = 0] &= 1 - x - y + I(x, y).
\end{aligned} \tag{7.7}$$

This is valid. Indeed, $I(x, y) \le x$ is equivalent to

$$y(1 + (1 - x)(1 - y)) \le 1, \tag{7.8}$$

which follows from

$$1 - y(1 + (1 - x)(1 - y)) = (1 - y)(x + (1 - x)(1 - y)) \ge 0. \tag{7.9}$$

Similarly $I(x, y) \le y$. Also,

$$I(x, y) \ge x + y - 1, \tag{7.10}$$

because, with $a = 1 - x$ and $b = 1 - y$, the difference is

$$(1-a)(1-b)(1+ab) - (1-a-b) = ab(2-a-b+ab) \geq 0. \tag{7.11}$$

Thus (7.7) has nonnegative entries and the correct marginals.

Implementing these couplings sequentially gives random sets whose marginals are uniform on $\mathcal{F}$. Let the corresponding unions be denoted by $A \cup B$ for the independent coupling and $A \cup C$ for the I-coupling. The chain rule and conditioning yield

$$\begin{aligned} H(A \cup B) &\geq \sum_i \mathbb{E}h(X_i Y_i), \\ H(A \cup C) &\geq \sum_i \mathbb{E}h(I(X_i, Y_i)), \end{aligned} \tag{7.12}$$

where X_i, Y_i have the same one-dimensional law as X_i. Applying Theorem 5 pointwise and averaging,

$$\mathbb{E}\big((1-\beta_*)h(X_i Y_i) + \beta_* h(I(X_i, Y_i))\big) \geq \frac{1}{m_*}\mathbb{E}\frac{X_i h(Y_i) + Y_i h(X_i)}{2}. \tag{7.13}$$

The right-hand side equals

$$\frac{\mathbb{E}X_i}{m_*}\mathbb{E}h(X_i) > \mathbb{E}h(X_i), \tag{7.14}$$

by (7.6). Summing over i gives

$$(1-\beta_*)H(A \cup B) + \beta_* H(A \cup C) > H(A). \tag{7.15}$$

But $\mathcal{F}$ is union-closed, so both unions lie in $\mathcal{F}$. Since A is uniform,

$$H(A \cup B) \leq \log|\mathcal{F}| = H(A), \qquad H(A \cup C) \leq H(A). \tag{7.16}$$

This contradicts (7.15). Therefore some element belongs to at least a c'-fraction of the sets. $\square$

A Computational certificates

This appendix records the finite certificates used in Sections 4 and 5. All interval computations use outward rounding and 50 decimal digits of precision. The implementation is available at

`https://github.com/kaare85-eng/liu-entropy-bound-certificate`. (A.1)

Proposition 7 (Constant enclosures)**.** *The constants x_*, m_*, β_*, x_0, u_f, and v_f are enclosed in intervals of radius less than 10^{-14} around the values displayed in* (2.9), (2.12), (5.36), *and* (5.37). *The enclosure for x_* certifies the unique root of* (2.5) *in* $(0, 1)$, *and the other constants are evaluated from their defining formulas.*

Proposition 8 (Bernstein positivity). *For*

$$P = P_0 + m_* P_1 + m_* \beta_* P_2,$$

all 28 degree-27 Bernstein coefficients of P on $[0,1]$ are positive. The minimum certified lower bound is

$$\min_i b_i > 4.550899799454681. \tag{A.2}$$

Consequently,

$$P(x) > 0 \qquad (0 \leq x \leq 1).$$

Proposition 9 (Diagonal signs). *The interval evaluation certifies*

$$\begin{aligned} &M(m_*) > 0, \quad M'(m_*) < 0, \quad M''(m_*) > 0, \quad M^{(3)}(m_*) > 0, \quad M^{(4)}(m_*) < 0, \\ &M''(x_*) > 0, \quad M''(0.1) < 0. \end{aligned} \tag{A.3}$$

The numerical margins are those displayed in (4.12), (4.14), *and* (4.26).

Proposition 10 (Unique bifurcation). *Let*

$$H(u) = \frac{1}{2}\Gamma_{vv}(u,0). \tag{A.4}$$

Then

$$H(u) = L(u) + \frac{1}{2(1-u)} - m_*\alpha_* L(u^2) - 2m_*\beta_*(1-u+u^2)L(D(u)). \tag{A.5}$$

The interval proof certifies

$$\begin{aligned} H(u) < 0 &\qquad (0 < u < x_0), \\ H(u) > 0 &\qquad (x_0 < u < 1), \end{aligned}$$

and a unique zero at x_0. On the root box, $H'(u) > 0$.

Proposition 11 (Fold uniqueness and value). *The system*

$$F(u,v) = 0, \qquad \Gamma_{vv}(u,v) = 0 \tag{A.6}$$

has a unique solution in the fold box centered at (u_f, v_f). At this solution,

$$\Gamma_u(u_f, v_f) > 0.41. \tag{A.7}$$

Moreover, the interval evaluation gives

$$\Delta(u_f, v_f) < -31.17. \tag{A.8}$$

Proof of branch isolation. We now explain how the preceding certificates are assembled into the branch isolation statement. Consider the zero set

$$\{(u,v) : F(u,v) = 0,\ v > 0\}$$

inside the feasible triangle

$$0 < u < 1, \qquad 0 < v < \min(u, 1-u).$$

Any connected component of this zero set must fall into one of the following topological alternatives.

First, it may be a closed loop in the interior. Such a loop must have at least two extrema in the u-projection, and hence at least two fold points, that is, at least two solutions of

$$F(u,v) = 0, \qquad F_v(u,v) = 0.$$

On $F = 0$ this is equivalent to

$$F(u,v) = 0, \qquad \Gamma_{vv}(u,v) = 0.$$

Proposition 11 certifies that there is only one such fold. Thus closed loops are excluded.

Second, a component may meet the diagonal $v = 0$. Since Γ is even in v, branches can bifurcate from the diagonal only where

$$\Gamma_{vv}(u,0) = 0.$$

Proposition 10 certifies that the only such point is $u = x_0$. Thus the only possible diagonal birth is the already traced lower branch.

Third, a component may approach the boundary of the feasible triangle away from the corner $(x,y) = (1,0)$. The side $y = 0$, away from $x = 1$, is excluded analytically: for fixed $0 < x < 1$,

$$F(x,y) \to -\infty \qquad (y \to 0^+).$$

The side $x = 1$, away from $y = 0$, is also excluded analytically: for fixed $0 < y < 1$,

$$F(x,y) \to +\infty \qquad (x \to 1^-).$$

The reflected sides $x = 0$ and $y = 1$ do not occur as distinct sides in the half-triangle $v > 0$; they are obtained by symmetry.

Fourth, a component may approach the corner $(x,y) = (1,0)$. The upper arc has exactly this endpoint. The uniqueness of this approach is certified by a one-dimensional interval Newton band computation in the variable $s = y$. For $s \in [0.04, 0.08]$, the computation certifies a unique solution of $F(\varepsilon, s) = 0$, where $\varepsilon = 1 - x$, in the upper-branch tube. The eight s-bands have width 0.005; the maximal Newton contraction factor is 0.6296. For $0 < s < 0.04$, Lemma 2 gives the analytic endpoint estimate and identifies the same unique branch approaching the corner.

The two known arcs account for the unique diagonal bifurcation, the unique fold, and the unique corner approach. The alternatives above exhaust all possible connected components. Therefore there are no additional components of $F = 0$, $v > 0$, in the feasible triangle.

Proposition 12 (Branch isolation)**.** *The zero set*

$$F(u,v)=0, \qquad v>0,$$

in the feasible triangle (5.3) *consists of exactly the two arcs stated in Proposition 3. The lower arc runs from the diagonal bifurcation point* $(x_0,0)$ *to the fold* (u_f,v_f)*. The upper arc runs from the endpoint* $(x,y)=(1,0)$ *to the same fold.*

Proposition 13 (Lower arm determinant sign)**.** *On the lower arc of* $F=0$*, the Taylor-model interval certificate verifies*

$$\Gamma_{vv}<0, \qquad \Delta<0. \tag{A.9}$$

The bifurcation zone uses (u,w)*-coordinates with* $w=v^2$*. The Taylor-model cover contains* 23 *boxes in the near-bifurcation zone and* 56 *boxes in the compact lower zone. The largest certified upper bound for* Δ *is*

$$-0.00116054425855. \tag{A.10}$$

Proposition 14 (Upper arm determinant sign)**.** *On the compact upper arc with* $y\in[0.08,y_f]$*, Taylor-model interval arithmetic in the coordinates*

$$(\varepsilon,s)=(1-x,y)$$

verifies

$$\Gamma_{vv}>0, \qquad \Delta<0. \tag{A.11}$$

The cover uses 42 *adaptive boxes with maximum step size at most* 0.001*. The largest certified upper bound for* Δ *is*

$$-31.1897707236, \tag{A.12}$$

and the smallest certified lower bound for Γ_{vv} *is*

$$0.000394450122601. \tag{A.13}$$

Proposition 15 (Endpoint positivity)**.** *For the endpoint region* $0<y\le 0.04$*, Lemma 2 gives the analytic lower bound*

$$\Gamma_u \ge \frac{2m_*-1}{2}\log\frac{1}{y}>0. \tag{A.14}$$

At the transition point $y=0.04$*, the computed value is*

$$\Gamma_u=0.745513582988373\ldots, \qquad \frac{2m_*-1}{2}\log 25=0.377544881375125\ldots. \tag{A.15}$$

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